Assignment 3

Title:

Create a form with fields: name, email, gender, date of birth, and submit button.

Aim:

To design and implement an HTML form that collects user information such as name, email, gender, date of birth, and includes a submit button.

Problem Statement:

Create a simple HTML webpage that contains a form with the following input fields:

- Name (Text field)
- Email (Email field)
- Gender (Radio buttons)
- Date of Birth (Date picker)
- Submit button

Theory:

Introduction to HTML Forms:

An HTML form is used to collect user input. Forms are commonly used for registration, login, feedback, and data submission.

The <form> tag is used to create a form in HTML.

Syntax:

```
<form action="server_page" method="post">
   Form elements
</form>
```

Attributes of <form>:

- action → Specifies where the form data will be sent
- method → Specifies HTTP method (GET or POST)

Form Elements:

1. <input> Tag

The <input> tag is used to create input fields. It is an empty tag.

Common Input Types:

Type	Description
text	Single line text input
email	Email input with validation
radio	Radio button
date	Date picker
submit	Submit button

Attributes of <input> Tag

The <input> tag is an empty tag used to create input controls.

Attribute	Description
type	Specifies type of input
id	Unique identifier
name	Name used to send data to server
value	Default value
placeholder	Hint text inside field
required	Makes field mandatory
maxlength	Maximum characters allowed
checked	Pre-selects radio/checkbox
readonly	Makes field non-editable
disabled	Disables field

2. Text Field

```
<input type="text">
```

Used to enter name or other text information.

3. Email Field

<input type="email">

Validates email format automatically.

4. Radio Button

<input type="radio">

Used to select one option from multiple choices.

5. Date Field

<input type="date">

Provides a calendar to select date.

6. Submit Button

<input type="submit">

Used to submit form data.

Explanation of Program Code:

1. <!DOCTYPE html> declares HTML5 document.
2. <form> creates form section.
3. type="text" creates name input field.
4. type="email" ensures valid email format.
5. type="radio" allows selection of one gender.
6. type="date" provides calendar selection.
7. type="submit" creates submit button.
8. required attribute ensures field must be filled before submission.

Working of the Program:

1. Save file as form.html.

2. Open in browser.
3. User enters required details.
4. Email field checks valid email format.
5. After clicking submit, form data is processed (currently # means no server).

Advantages:

- Collects user information efficiently
- Provides built-in validation (email type)
- Easy to design and implement
- Widely used in web applications

Applications:

- Registration forms
- Login forms
- Feedback forms
- Online admission forms
- Job application forms

Result:

The HTML form with name, email, gender, date of birth, and submit button was successfully created and executed.

Conclusion:

From this experiment, we learned:

- How to create forms using <form> tag
- How to use different input types
- How to use radio buttons
- How to use date picker
- How to submit form data

Thus, the form was successfully designed using HTML.

Questions & Answers:

1. What is the use of <form> tag?

2. What is the difference between GET and POST method?

3. What is the use of required attribute?

4. What is radio button used for?

5. Which input type validates email automatically?